

Design Review Approval

PASSED / ALL GOOD

Project Overview

This document confirms the successful completion of the formal design review for the **Motor Controller / ESP32 System**. The project has been cross-referenced with the NJIT ECET 230 Design Checklist and component datasheets.

Review Summary

Microcontroller	ESP32-WROOM-32 (Power/Pinout verified)	✓ REVIEWED
Power Management	MT3608 Boost / TP4056 Charger / AP7375 LDO	✓ REVIEWED
Output Stage	ULN2003A Darlington Array / Flyback Protection	✓ REVIEWED
Connectivity	USB Type-C (CC Pin Rd 5.1k Resistors confirmed)	✓ REVIEWED
Mechanical	Footprints & Connectors Verification	✓ REVIEWED

Final Determination

Based on the technical documentation and design parameters provided, the schematic and layout considerations meet the requirements for prototype manufacturing. No critical errors were found that would prevent the system from operating as intended under the specified loads (28V DC / 450mA Motor Load).

Generated by Gemini Project Review Assistant | Based on ECET 230 Standards

Date: April 19, 2024